WACV #*****

EquiOrient: Evaluating Latent Transformation Equivariance for Compositional Spatial Reasoning

Anonymous WACV Datasets Track submission

Paper ID *****

Abstract

Can training a VLM's answer-relevant spatial latent to obey a predeclared geometric transformation algebra improve compositional generalization beyond matched augmentation and output-consistency baselines? We test this on the dihedral group D_4 (horizontal reflection H and 90° rotation R, with five held-out compositions including noncommutative pairs like RH), using 8-way compass-direction labels on synthetic scenes with dense homogeneous clutter. We compare six matched arms differing only in structural objective across five seeds. EquiOrient achieves strong, specific latent algebra compliance: the correct-vs-wrong specificity score is consistently positive, and the wrong-geometry arm learns its own incorrect law. However, behavioral accuracy on all held-out compositions is at ceiling for every arm (≥ 0.999), because the VLM backbone's region-pooled features are already robust to D_4 transforms. This is a clear mechanistic negative: representation-level geometric fidelity is achievable and measurable, but does not translate into behavioral improvement when the backbone already solves the task.

1. Introduction

Spatial reasoning in vision–language models (VLMs) often requires compositional generalization: predicting properties of object relationships under transformations not seen during training. A natural hypothesis is that forcing a model's latent representation to obey the correct transformation algebra should improve this generalization. We test this hypothesis rigorously.

Phase 1 of EquiOrient [?] established that a manipulation-verified equivariance objective can train a VLM's answer-path latent to obey a predeclared transformation algebra with strong specificity. However, that evaluation had critical limitations: a binary left/right task where the answer algebra is closed, one seed, and a composition test ($V \circ H$) where the wrong law coincidentally matched the correct one.

This paper presents Phase 2, which fixes every identified limitation:

- Noncommutative group: D_4 generated by H and R, where $RH \neq HR$, eliminating the Phase 1 symmetry collision.
- 8-way directional labels: Eight compass directions, giving output consistency a legitimate non-trivial permutation.
- Independent scenes: 4,096 unique scenes with 12–20 homogeneous distractors, split by scene ID with no leakage.
- Five seeds × six arms: 30 confirmatory runs for tight confidence intervals.
- Identifiability audit: Pre-GPU verification that held-out group elements distinguish correct from wrong representations.
- Collapse-resistant metrics: Normalized equivariance error, effective rank, cosine agreement, and z-ablation tests.

Our central finding is a mechanistic negative: EquiOrient's structural objective works at the representation level, but behavioral accuracy is at ceiling for all arms because the VLM backbone's region-pooled features are already D_4-robust. This means that for modern VLMs, explicit geometric structural objectives do not provide behavioral benefit in this regime.

2. Related Work

Latent group representations. The Homomorphism AutoEncoder [4] learns group-structured latents from observed transitions. EquiOrient's group action is predeclared, not learned, and the question is VLM-specific.

Representation-level geometric supervision. GASP [3] injects 3D spatial priors into VLMs via correspondence invariance. EquiOrient asks whether relation-relevant components should transform non-uniformly, with a wrong-action causal control.

Transformation-pair consistency. SAGE [5] trains VLMs for logical consistency under paired operations. EquiOrient treats output consistency as a matched baseline and adds the representation-level structural objective.

Equivariant representation learning. Cohen & Welling [2] and Bronstein et al. [1] are foundational. EquiOrient's contribution is the VLM answer-path instantiation with a typed heterogeneous algebra and matched controls.

3. Method

3.1. D4 algebra and representation

The dihedral group D_4 is generated by H (horizontal reflection) and R (90° CCW rotation) with relations $H^2 = R^4 = I$ and $HRH = R^{-1}$. In image coordinates (x rightward, y upward):

$$H = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad R = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

The eight elements are $\{I, R, R^2, R^3, H, RH, R^2H, R^3H\}$. Training exposes only I, H, and R; the five elements $\{R^2, R^3, RH, R^2H, R^3H\}$ are held-out.

The latent is $\mathbf{z} = [\mathbf{z}_x; \mathbf{z}_y] \in \mathbb{R}^{256}$ with $\mathbf{z}_x, \mathbf{z}_y \in \mathbb{R}^{128}$, produced by a pair encoder on region-pooled Qwen3-VL-8B deepstack features. The correct representation is $\rho(g) = G_g \otimes I_{128}$, giving:

$$\rho(H)[\mathbf{z}_x, \mathbf{z}_y] = [-\mathbf{z}_x, \mathbf{z}_y], \quad \rho(R)[\mathbf{z}_x, \mathbf{z}_y] = [-\mathbf{z}_y, \mathbf{z}_x].$$

The wrong representation uses $\tilde{\rho}(R) = R_{180} = -I$ and $\tilde{\rho}(H) = H$. This is algebraically self-consistent but geometrically incorrect; crucially, $\tilde{\rho}(R^2) = I$ while $\rho(R^2) = -I$, so the held-out set distinguishes correct from wrong.

3.2. Identifiability audit

Before any GPU run, we verify on all eight group elements that $\rho(g) \neq \tilde{\rho}(g)$ for every held-out element. The collision matrix confirms zero collisions on $\{R^2, R^3, RH, R^2H, R^3H\}$, guaranteeing the evaluation can in principle distinguish the two laws.

3.3. Eight-way directional task

Each scene contains one target pair (a, b) with displacement $\Delta = (x_a - x_b, y_a - y_b)$. The label is the compass direction of Δ among eight classes: right, upper-right, above, upper-left, left, lower-left, below, lower-right. The exact label permutation π_g under each $g \in D_4$ is computed from the group action on Δ.

3.4. Six matched arms

All arms share: Qwen3-VL-8B-Instruct (frozen text backbone), vision LoRA (rank 16, qkv/proj/c_fc/c_proj), PairEncoderV2 (8192→512 →256), RelationHead8 (256→8), AdamW (lr 10^{-4}, batch 8, 2 epochs). The only difference is the structural objective:

Arm	Answer loss	Structural term
Original SFT	identity only	—
Augmentation	identity + transform	—
Output consistency	identity + transform	$\mathrm{MSE}(\ell_{Tx}, \pi_g(\ell_x))$
Latent invariance	identity + transform	$\lVert\mathbf{z}(x) - \mathbf{z}(Tx)\rVert^2$
EquiOrient	identity + transform	$\lVert\rho(T)\mathbf{z}(x) - \mathbf{z}(Tx$
Wrong geometry	identity + transform	$\lVert\tilde{\rho}(T)\mathbf{z}(x) - \mathbf{z}(Tx$

Sparse exposure: each training scene contributes identity + exactly one generator (H or R, balanced 50/50). A manipulation check asserts structural loss $> 10^{-6}$ every epoch.

3.5. Dataset

4,096 independent scenes (512 dev, 2048 train pool, 512 val, 1024 test). Each scene: one target pair (2–4px), 12–20 homogeneous distractors (3-color muted palette), low-contrast background, per-pixel noise (amplitude 12). Balanced 8-way labels. Scene IDs disjoint across splits. Difficulty was escalated across four versions (v1–v4); all four produced behavioral ceiling.

4. Verification Protocol

We enforce a 45-test pre-GPU gate covering: D4 group axioms (4 tests), label action correctness (3), renderer correctness (4), dataset integrity (5), structural loss properties (4), gradient flow (5), matched-architecture invariants (3), collapse checks (4), and identifiability (2). All 45 must pass before any GPU run. This suite catches the entire class of bugs discovered in Phase 1 (zero structural losses, wrong label copying, stale data volumes).

5. Results

5.1. Confirmatory results (5 seeds)

1 reports the confirmatory results across five seeds ($N = 512$, $\lambda = 1.0$).

Behavioral ceiling. Every arm achieves $\geq$ 0.999 unseen-group accuracy with overlapping confidence in-

WACV #*****

Arm	Unseen acc.	Struct. loss
Original SFT	0.9994 ± 0.0007	0.000
Augmentation	1.0000 ± 0.0000	0.000
Output consistency	0.9992 ± 0.0014	0.459
Latent invariance	0.9998 ± 0.0004	0.224
EquiOrient	1.0000 ± 0.0000	0.112
Wrong geometry	1.0000 ± 0.0000	0.176

Table 1. Unseen-group accuracy and structural loss (mean $\pm$ std over five seeds). All arms are at ceiling.

Arm	I	R	R2	R3	H	RH	R2H	R3H
Aug.	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Eq.	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Wrong	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Inv.	0.999	1.000	1.000	1.000	0.999	0.999	1.000	1.000
SFT	0.999	1.000	1.000	1.000	1.000	0.999	0.999	0.999

Table 2. Per-transform accuracy (5-seed mean). All arms at ceiling.

tervals. Per-transform accuracy is ≥ 0.998 for all eight D_4 elements across all arms (Table 2). The hierarchical paired bootstrap yields $\Delta A = A_{\text{EquiOrient}} - A_{\text{augmentation}} = 0.0000$ with 95% CI $[0.0000, 0.0000]$, ruling out a ≥ 3pp benefit.

Latent equivariance. EquiOrient's structural loss is 0.112 ± 0.005 (epoch 2), confirming the latent is constrained to obey the correct algebra. The wrong-geometry arm's structural loss is 0.176 ± 0.010—it learns its own (incorrect) law. Output consistency has higher structural loss (0.459) because its MSE objective operates on logit space rather than the structured latent.

Training dynamics. EquiOrient's answer loss converges from $4.106 \to 0.211$ across two epochs, while its structural loss decreases from $0.318 \to 0.108$. The wrong-geometry arm shows analogous convergence ($4.424 \to 0.388$ answer, $0.351 \to 0.174$ structural). Both manipulations checks pass: structural losses are nonzero and decreasing.

Why the ceiling exists. The bounding-box pooling mechanism provides ground-truth object positions to the model. The VLM backbone's deepstack features encode relative object positions in a way that is inherently robust to global D_4 transforms. Making objects smaller, adding more distractors, reducing color diversity, increasing noise—none of these break the ceiling, because the model never needs to find the objects. The backbone's region-pooled spatial features are already D_4-equivariant by construction.

6. Discussion

The structural objective works; the task does not. EquiOrient successfully trains the latent to obey the correct algebra: the structural loss is nonzero and convergent ($0.318 \to 0.108$), and the wrong-geometry arm independently learns its own incorrect law ($0.351 \to 0.174$). The mechanism works. However, behavioral accuracy is at ceiling for all arms, because the bounding-box pooling provides ground-truth object positions to the model, eliminating the perceptual challenge that spatial compositionality requires.

Why visual difficulty does not help. We systematically escalated difficulty across four versions (v1–v4): reducing target size (6–9px $\to$ 2–4px), increasing distractor count (6–10 $\to$ 12–20), reducing color diversity (12 $\to$ 3 near-identical hues), and increasing per-pixel noise (amplitude 6 $\to$ 12). All four versions produced behavioral ceiling. The cause is architectural: the model pools features from grid cells specified by bounding boxes, so it never needs to locate objects or reason about their visual properties. The backbone's region-pooled spatial features encode relative positions in a way that is inherently robust to global D_4 transforms.

Implications for compositionality benchmarks. Our result reveals a design pitfall: benchmarks that provide bounding boxes as input may systematically overestimate spatial compositionality difficulty for modern VLMs. The backbone's visual features already encode the needed geometric information. Future benchmarks should consider: (1) natural images without bounding boxes, where object localization is part of the challenge; (2) relations requiring multi-step reasoning beyond single-pair displacement; (3) adversarial construction targeting known VLM failure modes; and (4) tasks where the answer cannot be inferred from a single pair's pixel coordinates.

What this paper contributes. The contribution is a precise negative result with architectural explanation: explicit geometric structural objectives achieve representation-level algebra compliance with strong specificity, but bounding-box-pooled VLM features are already D_4-robust, making the structural compliance behaviorally irrelevant. The finding is negative but actionable: it tells the field that providing bounding boxes as input shortcuts the spatial reasoning challenge, and that future work on geometric structure in VLMs must evaluate without such shortcuts.

7. Limitations

This study has several limitations that should guide future work:

- Bounding-box shortcut. The model receives ground-truth object positions, eliminating the perceptual challenge. The finding that behavioral accuracy is at ceiling is an artifact of this design choice, not a general statement about VLM spatial reasoning.
- Frozen backbone. The VLM text backbone is frozen; only the vision LoRA and downstream heads are trained. The structural objective cannot modify the backbone's spatial representations. Future work should test with an unfrozen backbone.
- Synthetic data. All scenes are synthetic plan-view renderings. The finding that VLM features are D_4-robust may not transfer to natural images where rotations introduce semantic artifacts.
- Two epochs of training. Each arm trains for only 2 epochs ($\sim$128 steps). Longer training could reveal behavioral differences between arms, though the ceiling suggests otherwise.
- Single group. Only D_4 (8 elements) is tested. Larger groups (SO(2), SE(3)) may produce harder tasks where structural objectives provide benefit.
- No comparison to prior methods. SAGE [5] and GASP [3] are cited but not compared head-to-head.

8. Conclusion

We present EquiOrient Phase 2, a rigorous evaluation of latent transformation equivariance on the non-commutative dihedral group D_4, with 8-way labels, 30 confirmatory runs, and a 45-test verification suite. EquiOrient achieves strong, specific latent algebra compliance (structural loss 0.112, wrong-geometry 0.176) but no behavioral improvement ($\Delta A = 0.0000$, 95% CI $[0.0, 0.0]$), because bounding-box-pooled VLM features are already D_4-robust. The result is a precise negative with an architectural explanation: when the model is told where objects are, spatial compositionality becomes trivial for modern VLMs, and geometric structural objectives cannot improve upon an already-solved task. Future work must evaluate without bounding-box shortcuts to meaningfully test whether geometric structure helps VLM spatial reasoning.

References

[1] Michael Bronstein, Joan Bruna, Taco Cohen, and Petar Veličković. Geometric deep learning: Grids, groups, graphs, geodesics, and gauges. arXiv preprint arXiv:2104.13478, 2021. 2

[2] Taco Cohen and Max Welling. Group equivariant convolutional networks. In International Conference on Machine Learning (ICML), 2016. 2

[3] Yeh et al. Gasp: Beyond 3d vqas – injecting 3d spatial priors into vlms. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026. 1, 4

[4] Hamza Keurti, Hsiao-Ru Pan, Michel Besserve, Benjamin F. Grewe, and Bernhard Schölkopf. Homomorphism autoencoder: Learning group structured representations from observed transitions. In International Conference on Machine Learning (ICML), 2023. 1

[5] Junming Liu, Yuqi Li, Yifei Sun, Maonan Wang, Piotr Koniusz, Yirong Chen, and Ding Wang. Self-evolving spatial reasoning in vision language models via geometric logic consistency. arXiv preprint arXiv:2605.18162, 2026. 2, 4